Acid reactions

SEE ALSO

❮ **28–29** Respiration
❮ **89** Acid rain
❮ **128–129** Chemical reactions
❮ **144–145** Acids and bases
Alcohols **160–161** ❯

ACIDS REACT WITH A RANGE OF OTHER SUBSTANCES IN PREDICTABLE WAYS.

Although acids come in many forms, they all react in the same way. When any acid is added to metals, oxides, or other compounds, the reaction generates the same set of products.

Acids and metals

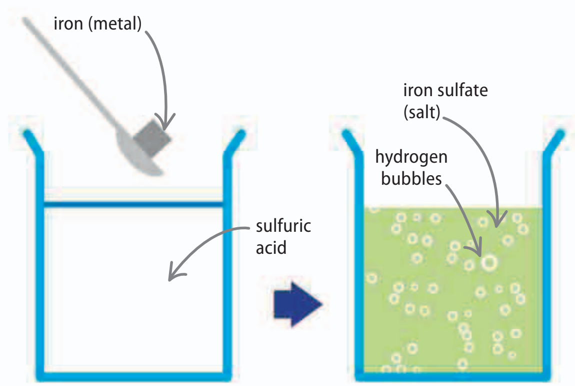
If a metal is more reactive than the hydrogen in an acid, they will react to form a salt and hydrogen gas. The most reactive metals, such as potassium, even do this with water—which contains hydrogen but is, by definition, neutral. Metals such as copper and gold are less reactive than hydrogen, so they do not react with most acids.

▽ **General equation**
Iron displaces the hydrogen in the sulfuric acid (H_2SO_4), forming a salt, iron sulfate ($FeSO_4$), which is an ionically bonded compound. The hydrogen has nothing else to react with, so it is released as a gas.

acid + metal = salt + hydrogen

$$H_2SO_4(aq) + Fe\,(s) \rightarrow FeSO_4(aq) + H_2(g)$$

▽ **Iron plus sulfuric acid**
When solid iron is added to the acid, the mixture begins to fizz with hydrogen bubbles. The iron sulfate salt dissolves forming a green solution.



REACTION OF METALS

Name	Reacts with water	Reacts with most acids	Level of reaction
potassium	yes	yes	high
sodium	yes	yes	
lithium	yes	yes	
calcium	yes	yes	
magnesium	no	yes	
aluminum	no	yes	
zinc	no	yes	
iron	no	yes	
tin	no	yes	
lead	no	yes	
copper	no	no	
mercury	no	no	
silver	no	no	
gold	no	no	no reaction

REAL WORLD

Acid rain

Rainwater is naturally slightly acidic because carbon dioxide gas in the air dissolves in it, making weak carbonic acid. When this acidic rain falls on certain stones it reacts with the chemicals in the stones, gradually eating away at them in a process called weathering.